Gaurang Parkar

Oslo, Norway | gaurang@gaurangparkar.com | gaurangparkar.com

Profile

Physics PhD with experience evaluating AI-generated Python and C++ code, conducting computational research, and communicating technical reasoning. Experienced with large-scale simulations and high-performance computing, with a background in teaching and engineering.

Experience

Independent Consultant Jan 2026–Present
Remote, Mercor Intelligence

- Solve and evaluate advanced problems across multiple domains, emphasizing correctness, clarity, and rigor.
- Review, validate, and improve solutions generated by AI models or other experts.
- Provide detailed reasoning and step-by-step derivations, assessing consistency and assumptions.
- Analyze and benchmark AI-generated C++ and Python code.

Research Fellow, University of Stavanger Sept 2019–June 2024

- Conducted computational research in quantum field theory, investigating the static quark–antiquark potential across confined and deconfined phases.
- Ran large-scale high-performance Monte Carlo simulations, using computing resources of up to 1,000 GPUs and 1,000 CPU cores.
- Published research on full-QCD and quenched-lattice studies; contributed to the DeepRTP grant funded by the Research Council of Norway.

Teaching Assistant, Stony Brook University 2018–2019

- Taught introductory mechanics and electromagnetism laboratory courses.

Teaching Assistant, San Francisco State University 2015–2017

- Taught introductory mechanics and electromagnetism laboratory courses.

Trainee Engineer, Mahindra and Mahindra Limited 2014–2015
Kandivali, Mumbai

- Worked in the tractor engine assembly plant.

Education

PhD in Physics, University of Stavanger 2019–2024
Thesis: *In-medium static quark anti-quark potential from lattice QCD*

Graduate Program in Physics, Stony Brook University 2017–2019
Research project: Tensor network methods in quantum information theory

MSc in Physics, San Francisco State University 2015–2017
Thesis: *Bound on Chaos in 2D CFTs*

BTech in Chemical Engineering, National Institute of Technology, Tiruchirappalli 2010–2014

Skills

Programming: Julia, Python, Bash scripting, C, C++, Wolfram Mathematica
Computing and software: High-performance computing, parallel processing (CPU and GPU), GNU/Linux, Windows, macOS, Git, Visual Studio Code, LaTeX, Microsoft Office
Scientific: Statistics, machine learning, statistical mechanics, thermodynamics, stochastic simulations, linear algebra, calculus, quantum field theory, quantum information and computing, electrodynamics, fluid dynamics
Languages: English (full professional competence); Norwegian (A2–B1)

Selected Publications

- R. N. Larsen, G. Parkar, A. Rothkopf, and J. H. Weber, "In-medium static inter-quark potential on high resolution quenched lattices," *Physical Review D*, 110(11), 114501, 2024. DOI
- D. Bala et al., "Static quark-antiquark interactions at nonzero temperature from lattice QCD," *Physical Review D*, 105(5), 054513, 2022. DOI
- G. Parkar et al., "Complex potential at $T > 0$ from fine lattices," *Proceedings of Science*, LATTICE2022, 188, 2023. DOI

Full publication list: Google Scholar